

Introduction for students: why Chapter 3 matters

Chapter 3 turns the laws of Chapter 2 into shortcuts. Instead of writing a KCL or KVL equation for every node and loop, you will combine resistors, split a voltage or a current in one line, and handle networks that look impossible at first sight. Most circuits in this course, and most of the midterm, use these shortcuts.

What you will learn

- **Series resistors:** they carry the same current, and their resistances add.
- **Parallel resistors:** they share the same two nodes and the same voltage. For two of them, the equivalent is the product over the sum, $R_1 R_2 / (R_1 + R_2)$.
- **Voltage division:** in a series string, each resistor takes a share of the voltage in proportion to its resistance.
- **Current division:** in a parallel group, each resistor takes a share of the current in inverse proportion to its resistance.
- **Loading:** connecting a load to a voltage divider changes its output, unless the load is much larger than the resistor it sits across.
- **Delta-to-wye (Δ -to-Y) and wye-to-delta (Y-to- Δ):** the tool for bridges and other networks with no series or parallel pairs.

Later in this course

Chapter	Where Chapter 3 comes back
4 Analysis techniques	Thévenin and Norton equivalents, source transformations and maximum power transfer all need R_{eq} ; dividers are the quickest way to check a node voltage
5 Operational amplifiers	The gain of an inverting or non-inverting amplifier is set by a ratio of two resistors; in the non-inverting amplifier they form a voltage divider
6 Inductors and capacitors	Inductors combine like resistors; capacitors combine the other way round
7 First-order circuits	The time constant $\tau = RC$ or L/R uses the equivalent resistance seen by the capacitor or inductor

In your program

- **Electrical and Electronics:** a transistor amplifier is biased with a voltage divider, and every sensor input is a divider whose loading must be checked.
- **Computer, Communication and Telecom:** pull-up resistors and level shifters are voltage dividers, and sensors are often connected to a microcontroller's analog input through one.
- **Biomedical:** strain gauges and many pressure sensors sit in a Wheatstone bridge, the circuit you will analyse in S12 and S13.
- **Mechanical and Industrial:** a heater with two elements gives low, medium and high settings by connecting them in series, alone or in parallel.
- **Surveying:** the level sensors in many instruments are read through a bridge or a divider, so their output depends on the same ratios.

As an engineer

- **One line instead of four equations.** A divider gives a voltage or a current in one step, and it tells you at a glance how the answer changes when a resistor changes.
- **Loading is a design rule.** A divider only delivers its design voltage if the load is much larger than the bottom resistor (S4, S6, and Nilsson Example 3.4).
- **Bridges measure.** When a bridge is balanced, no current flows across the middle; that is how a Wheatstone bridge measures an unknown resistance precisely (S13).

In real life

- **Your generator cable.** A neighbourhood generator may hold 220 V at its terminals, but the cable to your home has resistance. The cable and your appliances form a voltage divider, so your home receives less than 220 V, and more so when you draw more current (S15).
- **A resistive touch screen.** Screens of this kind, still used on some card terminals and industrial panels, find your finger with one voltage measurement per direction: the touch splits a resistive sheet into two parts, and the measured voltage is a fraction of the supply set by where you touch (S16, and Nilsson's Practical Perspective for this chapter).

How to study this chapter, and where AI fits

- AI tools are allowed on the practice sheet. Use them to check your work, not to replace it.
- In a 2025 study of Gemini 2.5 Pro on undergraduate circuit problems, misread source polarities caused about a third of its wrong answers, and misread current directions nearly as many (arXiv 2512.10159)¹. In this chapter, the commonest slips are putting the wrong resistance in a divider formula and treating two resistors as series or parallel when they are not. S9 shows one: find it before you trust any answer.
- Check every answer yourself: resistors in parallel must have the same voltage, a divided voltage or current can never be larger than the total, and the powers must add to zero. These checks need no device.
- The quiz is device-free, and every question is a twin of a practice-sheet question. If you can do the sheet on your own, you will do well on the quiz.

Checklist: you can do these without help

- ☐ Say which resistors are in series and which are in parallel, and why
- ☐ Spot a wire that shorts out a resistor
- ☐ Find the equivalent resistance of a series-parallel network
- ☐ Use voltage division and current division, and check the result
- ☐ Explain why a load changes a divider's output
- ☐ Convert a Δ to a Y and a Y to a Δ , and use it to solve a bridge

¹<https://arxiv.org/html/2512.10159>